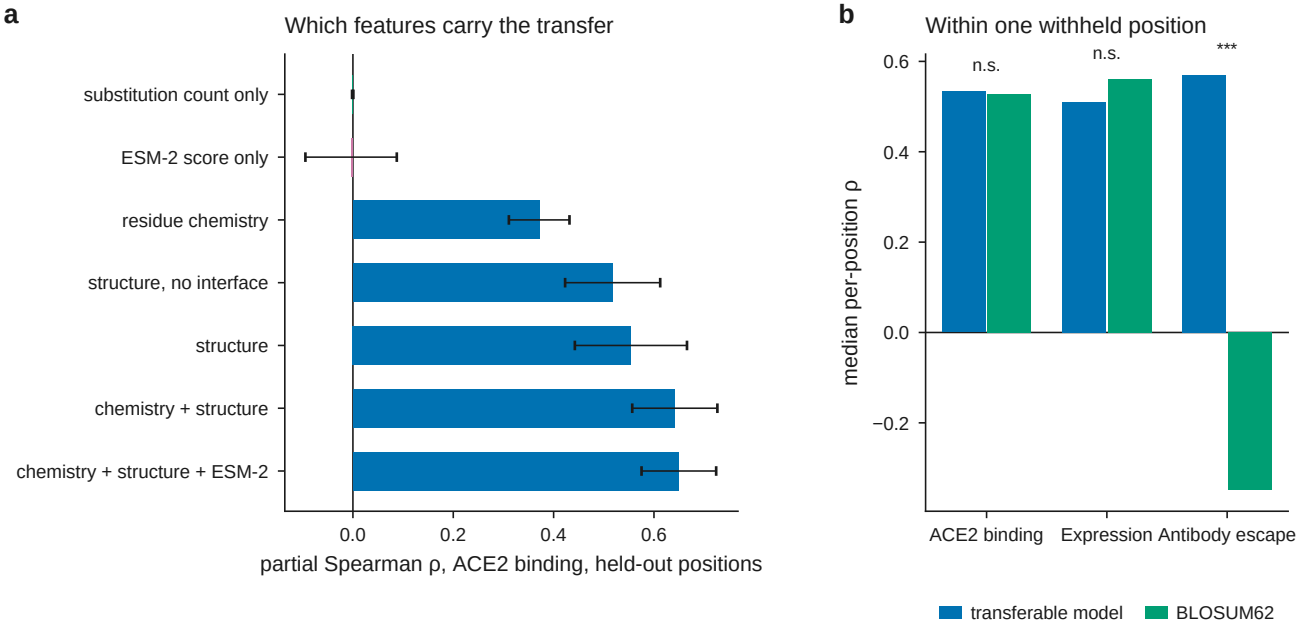
a
Which features carry the transfer
substitution count only
ESM-2 score only
residue chemistry
structure, no interface
structure
chemistry + structure
chemistry + structure + ESM-2
0.0
0.2
0.4
0.6
partial Spearman ρ, ACE2 binding, held-out positions
b
Within one withheld position
median per-position ρ
0.6
0.4
0.2
0.0
−0.2
n.s.
n.s.
***
ACE2 binding
Expression
Antibody escape
transferable model
BLOSUM62